

MHF4U - Advanced Functions

Assignment 2 - Polynomial and Rational Functions

Student name 	Date
Total marks 50	Suggested due date September 25, 2026 at 11:59 p.m. ET

- use factoring and theorems to analyze polynomial functions
- distinguish holes, restrictions, and asymptotes in rational functions
- build and critique mathematical models with stated limitations

Submission instructions

Factor completely before interpreting graph features. Keep restrictions from the original expression. Include labelled sketches where requested and check candidate solutions in the original equation.

Submit one readable PDF through the matching Moodle assignment. Include your name, page numbers, and complete reasoning. Do not submit shared links or temporary draft-file URLs.

1. Analyze $P(x) = -(x + 2)(x - 1)^2(x - 4)$. State degree, leading coefficient, zeros, multiplicities, end behaviour, and y-intercept. **/8**

2. Use the factor theorem to determine whether $x - 3$ is a factor of $2x^3 - 5x^2 - 4x + 3$. Explain the conclusion. **/6**

3. Factor $x^3 + 2x^2 - 9x - 18$ and use the result to sketch a graph with all intercepts and end behaviour labelled. /8

4. For $R(x) = (x^2 - x - 6)/(x^2 - 4)$, state restrictions, any hole, vertical asymptotes, horizontal asymptote, and intercepts. /10

5. Solve $2/(x - 1) = 3/(x + 2)$. State restrictions first and verify the final solution. /6

6. A rectangular open-top display tray has volume $V(x) = x(30 - 2x)(22 - 2x)$. State a realistic domain and explain two limitations of this polynomial model. /6

7. Compare the graph behaviour at a polynomial zero of even multiplicity with a removable discontinuity in a rational function. /6

Completion check

Before submitting: confirm every restriction is stated, graphs are labelled, units are included, and the total PDF is readable at 100% zoom.